Semantic aware receiver scheduling

System design algorithms simulation and development results

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Our goal is to reduce how long important images wait at the receiver before processing begins. Multiple user devices send complete messages. The receiver selects one currently available message whenever its single processing server becomes free.

This report explains the semantic inputs, scheduling policy, waiting objective, training methods, simulated UE streams and available results. Fourteen learned methods have completed development evaluation. LambdaMART has the lowest reported penalty among those learned methods. The protected final test remains unopened.

The central research question is whether priority, mission relevance, novelty and scene evidence improve important image waiting beyond conventional scheduling and strong semantic rules. A larger model does not establish that benefit. Matched comparisons and parameter ablations must show it.

Scope and reading guide

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2	Semantic parameters and transport
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The timing convention is one simulation unit equal to one simulated millisecond. Simulated service durations are assumptions, not hardware measurements. Scores reported as waiting penalties remain dimensionless.

1 System boundary and policy

Each offered message contains semantic parameters and a complete opaque digital payload. Only complete messages become ready. The scheduler reads parameters and permitted context. It does not inspect images or latent contents.

At decision time t, the scorer evaluates every ready candidate i using s_i = f(candidate parameters, queue context, permitted history). It selects the candidate with the largest score. Candidate identity routes the selection but must not become a learned semantic feature.

Processing is nonpreemptive. Messages can arrive while the server works. They become candidates at the next decision. A new urgent arrival does not interrupt the current job. The scheduler takes a fresh snapshot after processing ends, rather than committing to a permanent sorted queue.

Step	Receiver behavior
1	Admit complete messages whose arrival time has been reached
2	Build an immutable ready queue and permitted history snapshot
3	Score the candidates with the selected policy
4	Dispatch exactly one ready message
5	Advance the simulated server while admitting arrivals
6	Record waiting time and processing outcome
7	Update permitted history and select again

One shot operation and history

Each message receives one attempt and no automatic retry. Successful history describes evidence processed earlier at the receiver. It does not imply a retransmission protocol. For an all success control, every completed job enters that history. Outcome stress tests can assign saved failures that consume service time without adding successful evidence.

The current environment retains up to 32 successful processed headers. Temporal windows contain up to three recent observations. The history consists of parameter records rather than reference labels or evaluator fact identities.

Delay is the objective rather than the policy

A policy chooses the next candidate. Delay measures the operational effect of that choice. A ranker, a tree and an RL agent can all optimize the same waiting objective through different learning procedures. This separation lets us compare learning mechanisms without changing what success means.

2 Parameters and transport

Input	What it contributes
SPI	Four uncalibrated producer scores for Critical High Medium and Low
Context relevance	Numeric alignment between the image and its UE mission
Novelty	Difference relative to the transmitter knowledge base before committing this observation
Scene evidence	Retained detections normalized largest box areas and availability
Status	Missingness cold start and detector status

The received vector contains 387 entries. Scene evidence contributes 374. Four SPI scores, relevance, novelty, six further availability indicators and one cold start indicator supply the remaining thirteen. Context builders derive queue and history representations from those entries.

Scene vocabulary

All 93 categories enter the current training representation. They include the 80 COCO objects, five extra objects and eight region categories. The extra objects are building, bridge, utility pole, traffic sign and animal. The regions are road, water, debris, smoke, flame, snow, ground and grass.

Counts apply to 85 object categories. Region counts remain unavailable. Area means the largest detected bounding rectangle divided by image area. It is not segmentation coverage or physical distance. A person detection does not establish distress. A building detection does not establish damage. Many objects do not establish urgency.

Novelty follows a causal transmitter sequence. Query the existing knowledge base, calculate novelty, serialize parameters, then commit the observation. The same offered stream determines transmitter state across policies. The current observation cannot enter its own comparison history.

Mission relevance is an extracted numeric input. The current student does not read the mission paragraph. High novelty can describe an unimportant new photograph. A model must interpret novelty with priority and mission evidence.

Communication cost

Scene representation	Bytes
Original dense	369
Compact dense	307
Compact sparse	29 plus four times the active category count
Ten active categories in sparse form	69

Compact encoding preserves the stored measurements and masks under a preinstalled producer and vocabulary profile. These values cover the scene block only. SPI relevance novelty routing and timing add overhead. One combined binary envelope size has not been established.

We have not measured which categories are necessary. A smaller vocabulary requires a retrained comparison between no scene input, a compact subset and all 93 categories on identical splits and streams.

3 Waiting objective and training references

Queue waiting time is dispatch time minus arrival time. Completion delay adds service duration. Our implemented D objective measures waiting until dispatch. A Low image entering an empty queue can legitimately wait less than a Critical image entering a burst. The goal concerns matched workload results rather than a strict class mean ordering in every episode.

Reference class	Budget in simulated ms
Critical	2
High	4
Medium	8
Low	16

These budgets are experimental assumptions. They are not published emergency response deadlines. The constant service control takes one simulated ms per job. Historical files retain their original simulation unit representation.

Penalty

The penalty is ell(w,b) = w/b + max(w/b - 1, 0). Waiting is w and the class budget is b. The slope doubles after the budget. For Critical images, waits of 1, 2 and 4 ms produce penalties of 0.5, 1 and 3. A Low image waiting 4 ms contributes 0.25.

The episode score J_D is the mean penalty across offered messages. Lower is better. An unselected message contributes elapsed waiting up to the observation horizon. The score is dimensionless and must not be presented as a mean number of milliseconds.

Supervision

The reference target is the extra penalty caused by waiting one additional service interval. It equals ell(w + 1,b) minus ell(w,b). Larger targets identify candidates whose postponement is more costly under the stated objective.

Choice models learn the best target candidates. Regression predicts target magnitudes. Pairwise losses learn their relative order. Training references remain separate from predicted runtime inputs.

The completed target depends on reference priority and waiting time. It does not independently reward every within class mission or novelty distinction. Scene evidence may improve reference priority prediction. Current results alone cannot prove a broader understanding of usefulness.

RL reward

The environment computes reward as the negative change in total accumulated waiting cost divided by 100. The factor scales the reward. It does not change the preference for lower waiting cost. The selected message stops accumulating queue waiting at dispatch while other messages continue waiting.

4 Conventional policies and learned scoring

Policy	Selection rule
FIFO	Earliest arrival
Round robin	Rotate among sources
Seeded random	Random ready candidate
SPI FIFO	Highest predicted class then arrival
Relevance	Highest context alignment
Novelty	Highest novelty
SPI novelty	Priority then novelty
Predicted EDF	Earliest arrival plus predicted class budget

These controls separate arrival order fairness chance and simple semantic signals. Predicted EDF uses assumed budgets derived from predicted classes. It does not represent real application deadlines.

Linear and MLP choice

Linear choice learns a shared linear score from contextual features. MLP choice uses hidden widths 64 and 32 with ReLU activations. Both apply softmax over ready candidates and choice cross entropy. Best target ties receive equal target probability. Completely tied snapshots do not impose an arbitrary choice label.

Delay regression and priority classifier

Delay Huber regression predicts the marginal penalty target with Huber loss and delta equal to one. This tests learning target magnitudes. The completed priority classifier uses four class cross entropy. It is not the earlier proposed binary useful next classifier.

RankNet

RankNet predicts pair preference with sigmoid of the score difference. Binary cross entropy trains that preference. Our implementation weights strict pairs by normalized target gap, gives smaller weight to ties and bounds sampled pairs per snapshot. This adapts the neural pairwise mechanism in Burges et al [1].

LambdaMART and listwise ranking

LambdaMART uses boosted trees to learn nonlinear tabular interactions. It is a strong comparison for numeric parameter inputs and follows the ranking family described by Burges [2]. The listmle arm uses a tie aware strict lower surrogate that penalizes lower target candidates outranking higher ones. It is a listwise adaptation rather than the original permutation likelihood objective [5].

5 Set and temporal models

Deep Sets RankNet

A shared encoder maps candidates to learned vectors. Mean pooling summarizes the queue. Another mean summarizes successful processed history. The scoring head combines the candidate, its encoding, pooled context, sizes and comparable distances. RankNet supplies the training objective.

The representation supports variable queue sizes and does not assign meaning to arbitrary array position. Candidate permutation must preserve identity associated scores. Deep Sets provides the shared encoder and invariant pooling foundation [3]. Our implementation uses mean pooling.

Attention based set ranker

The attention model uses a 32 dimensional projection and two Transformer encoder layers with four heads. Candidates can exchange information before scoring. Pairwise loss trains selection order. The registry calls this set_transformer_ranknet. It is an attention based adaptation rather than a literal reproduction of every module in Lee et al [4].

Temporal Deep Sets

A GRU encodes up to three recent observations in the temporal windows. The model combines those short ordered histories with candidate and queue context. This tests whether ordering adds value beyond pooled history. It does not establish long term event tracking or natural cross view matching.

How scene information can contribute

SPI supplies a coarse priority assessment. Relevance identifies mission alignment. Novelty compares current observations with earlier transmitter observations. Detected categories and areas supply content context. The scorer can learn combinations but cannot recover information absent from those parameters.

For two High candidates, zero relevance could mean an off mission image, an extractor error or important evidence outside the device mission. A large object count alone cannot resolve those cases. The supervision

must distinguish them. The current delay target is narrower than a full operational judgment of emergency usefulness.

Architecture and supervision are separate choices. Deep Sets defines representation. RankNet defines a loss. Attention defines interactions. RL learns through transitions and reward. A larger model cannot repair a target that omits the desired distinction.

6 Reinforcement learning and execution

Masked Double DQN

The online network selects the next legal candidate and the target network evaluates it for the TD target. Masks exclude messages outside the ready queue. Double Q learning separates action selection and evaluation to reduce action value overestimation [6]. Candidate conditioned scoring adapts the mechanism to a changing queue.

Prioritized replay

TD prioritized replay samples stored transitions according to prediction error. Importance weights account for changed sampling probability. Replay priority controls learning samples rather than operational message urgency [7].

Masked PPO

PPO learns an actor over ready candidates and a value estimate. The legal action mask prevents invalid selections. Its clipped objective limits the effect of large probability ratio changes during updates [8].

Ranker warmstarted PPO

A trained ranker initializes the PPO actor. PPO then updates it using delay rewards. The comparison asks whether supervised initialization helps within the same bounded interaction budget. This is our hybrid training adaptation.

Component	Completed scope
Supervised methods	64 development episodes and 512 sampled decision snapshots per method
Epoch limit	50 with patience eight
RL budget	20000 action decisions per method
Model seed	1729
Validation	15 traces across UE counts and loads
Managed runtime	Python 3.13.15 Torch 2.11.0 and two T4 GPUs per paired notebook

A decision is one selection rather than an episode. Cached evidence avoids image producer retraining. The paired runs lasted about 2 hours 56 minutes and 2 hours 46 minutes. These are execution totals for their assigned computation. The budgets do not establish convergence or stability across seeds.

7 Data and dynamic simulation

Dataset	Images	Role

Dataset	Images	Role
CrisisMMD	4000	Disaster imagery
Open Images	1000	Everyday object diversity
Places365	1000	Scene diversity

The collection has 6000 images and 5952 raw unique images or cached embedding entries. Full pool split counts are 5376 training, 417 validation and 207 test. The assessable delay reference cohort has 3915 records. The final test remains unopened.

CrisisMMD supports humanitarian image analysis [9]. Its published labels do not automatically supply four class scheduling urgency. Current priority references are silver labels rather than independent operational urgency truth. The everyday sources broaden inputs but do not supply assessable urgency labels for the completed D comparison.

Fifteen UE profiles

The device and mission templates are general phone, citizen incident phone, structural inspection drone, person search drone, flood drone, wildfire drone, road inspection drone, connected car camera, emergency vehicle camera, traffic CCTV, public space CCTV, river or underpass camera, industrial robot, agricultural drone and maintenance camera.

These are project templates rather than fifteen official 6G device classes. A normal phone can capture important evidence. A disaster drone can send a routine repeat. The device mission supplies context rather than automatic importance.

Traffic and stream construction

Development validation includes 4, 12, 50, 60 and 100 UEs. The rate study includes 12, 50, 60 and 100. Fixed total load separates population from aggregate demand. Fixed per UE rates allow demand to grow with population. Burst settings change concentration. Joining and pausing alter achieved traffic, so intended and realized rates both matter.

The simulator saves arrivals, service durations and potential outcomes before comparison. Every policy receives the same realization. The transmitter knowledge base follows the offered sequence independently of receiver choices. Mixed image sampling supplies diversity. Mission sampling uses image text similarity. Neither establishes natural incident correspondence.

Capture time, extraction readiness, receiver arrival, dispatch and completion are distinct. Generated metadata must be marked simulated. Realistic stream construction requires supported sequences rather than unrelated images assigned invented physical correspondence.

8 Available development results

The following fourteen methods completed development training and evaluation. Values are dimensionless J_D penalties. Lower is better. They are not mean waiting milliseconds.

Model	Validation penalty
LambdaMART	3.2183
Delay Huber regression	3.3573
RankNet	3.3642

Model	Validation penalty
Deep Sets RankNet	3.3819
Tie aware listwise	3.3859
Temporal Deep Sets	3.4480
Attention based set ranker	3.4520
MLP choice	3.5675
Ranker warmstarted PPO	3.6375
Priority classifier	3.8271
Linear choice	4.3392
Masked PPO	4.5935
Masked Double DQN	4.7099
Double DQN with TD replay	5.1757

LambdaMART has the lowest penalty among these learned models. Several supervised methods outperform the completed RL variants under the configured budget. Attention and short temporal history did not improve on the best simpler methods in this table. Warmstarted PPO performed better than the other completed RL methods.

This table does not establish improvement over SPI first. Its matched baseline score and class safeguards must be merged into the comparison. One seed cannot establish stable superiority. Aggregate penalty alone cannot identify which class benefited or lost.

Completed rate study

The CPU rate experiment produced 320 rows from 80 traces and four policies. It crossed population, fixed total or per UE traffic, burst and sampling settings. Policies were FIFO, round robin, SPI first and SPI novelty. It completed in about 44 seconds. Metrics were finite and event and class accounting conserved messages. This establishes a reproducible workload comparison rather than final scientific evaluation.

9 Evaluation and evidence boundaries

Class tables must show offered selected and unselected counts, mean and median selected waiting, restricted mean wait and fraction selected within budget. p95 requires at least twenty selected observations in the implementation. Selected only means need unselected counts beside them.

Restricted waits use a 16 ms cap for messages with sufficient follow up. A message remaining unselected contributes the cap. This prevents undelivered images disappearing from the delay analysis. Snapshot nDCG uses declared gains and fixed queue context. It is separate from rollout performance.

Evidence metrics use declared fact identities. Exact copies can share an identity and receive no extra distinct credit. This construction does not establish that natural camera views show the same building.

Validation requirements

Check empty queues, arrivals during processing, busy dispatch, duplicate and foreign completions, reset, nonfinite outputs and horizon censoring. Verify penalty arithmetic independently. Compare exported predictions and selected identities with the training backend. Keep reference labels, future arrivals and evaluator identities outside runtime observations.

Remaining comparisons

Merge baseline and learned results on identical traces. Report each priority class with misses. Retrain no scene, compact scene and full scene variants on matched groups. Measure policy latency and context construction separately with hardware queue size warmup and counting convention. FLOPs and measured latency are different quantities.

Final uncertainty must respect independent source or duplicate groups. Queue seeds alone are not independent disasters. Freeze model and comparison choices before opening the final test. Separate broad image pool coverage from the narrower assessable priority cohort.

Local evidence

The environment is Evaluation/semantic_learning_env.py. Learning models are Scheduler/training/semantic_learning_models.py. The vocabulary is parameters/scene_vocabulary.py. Transport is Evaluation/compact_scene_transport.py.

Development evidence is under experiments/master_study/runs/paired-launch-20261102-205933/checks/paired-check-20261102T220621Z. Rate outputs are under experiments/stream_scheduler/runs/gap-repair-20261102T221315Z/rate_cpu_full. The existing Kaggle map is parameters/KAGGLE_RUNS.md.

10 References

[1] Burges et al 2005 Learning to Rank using Gradient Descent

[2] Burges 2010 From RankNet to LambdaRank to LambdaMART

[3] Zaheer et al 2017 Deep Sets

[4] Lee et al 2019 Set Transformer

[5] Xia et al 2008 Listwise Approach to Learning to Rank

[6] Van Hasselt et al 2015 Double Q learning

[7] Schaul et al 2015 Prioritized Experience Replay

[8] Schulman et al 2017 Proximal Policy Optimization

[9] Alam et al 2018 CrisisMMD

These references support the stated architecture and learning mechanisms. They do not validate the selected waiting budgets, silver priority references or project performance. Those claims depend on implementation and experiment evidence.